AI Engineering Overview

Large Language Models are neural networks trained on vast amounts of text data. They learn to predict the next token in a sequence, which enables them to generate coherent and contextually relevant text.

Tokens and Context Windows

A token is roughly three quarters of a word. Claude has a 200K token context window, meaning it can process about 150000 words in a single request. You pay per input and output token.

RAG - Retrieval Augmented Generation

RAG is a technique that combines vector search with LLM generation. Documents are split into chunks, embedded as vectors, and stored in a vector database. When a user asks a question, the question is embedded and the most similar chunks are retrieved and passed to the LLM as context.

Embeddings

Embeddings are numerical representations of text. Similar text produces similar embeddings. Voyage AI produces 1024-dimensional embeddings. Cosine similarity measures how close two embeddings are, with 1.0 meaning identical and 0.0 meaning completely unrelated.

Vector Databases

Vector databases store embeddings and enable fast similarity search using Approximate Nearest Neighbor algorithms. Supabase pgvector combines Postgres SQL with vector search. You can filter by metadata like user_id alongside semantic search.

Prompt Engineering

Good prompts follow the Persona plus Constraint plus Format pattern. Temperature 0.1 gives near-deterministic output for JSON parsing. XML tags separate instructions from user content. Always define null cases to prevent KeyErrors.

Cost Optimization

Prompt caching reduces system prompt token costs by 90 percent. Batch processing handles multiple inputs in one API call. Model routing sends simple tasks to Haiku at 10x lower cost than Sonnet. Context compression reduces tokens sent per query by 40 to 60 percent.